

Nayanpreet Chhabra

+1 (607) 297 0785 || nayanpreet9261@gmail.com || [LinkedIn](#) || [Portfolio](#) || New York (Open to Relocation)

Data Engineer with 2+ years of experience building scalable pipelines, transforming data, and ensuring data quality. Skilled in SQL, Python, Spark, and modern platforms like BigQuery and Snowflake, with expertise in ETL/ELT and data modeling.

EDUCATION

Master of Science in Information Systems – 3.94/4.00

Expected May 2026

State University of New York at Binghamton – Binghamton University, USA

Relevant Coursework - Data Warehousing, Advanced Database Systems, Data Mining, Machine Learning, Applied Statistics

Bachelor of Technology in Computer Science – 8.3/10.00

2018 – 2022

Shri Vaishnav Vidyapeeth Vishwavidyalaya – Indore, India

TECHNICAL SKILLS

Programming and Query: Python, SQL, C++

Data Engineering: ETL/ELT, Data Pipelines, Data Modeling, Data Warehousing, Data Quality, Workflow Orchestration, Airflow, Spark, Snowflake

Databases: PostgreSQL, MySQL, Teradata SQL

Analytics and BI: Tableau, Power BI, Excel, Matplotlib, Seaborn

Cloud and DevOps: AWS (S3, EC2, Lambda), Google Cloud Platform (GCP), Docker, Kubernetes, GitHub Actions, CI/CD

Machine Learning (Supporting): scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers

Tools: Git, Linux, Jupyter Notebook, VS Code

PROFESSIONAL EXPERIENCE

Binghamton University, *Senior Research Aide* | New York, USA

Apr 2025 – Feb 2026

- Developed and maintained **SQL queries and reporting pipelines** to support academic data systems and operational analytics.
- Automated recurring reporting workflows using **Google Workspace APIs**, improving efficiency and reducing manual reporting effort.
- Managed backend **data integrity and validation** for datasets supporting **200+ courses**, ensuring accurate reporting and certification records.

Vodafone, *Data Engineer* | UK Delivery, Remote

Aug 2022 – May 2024

- Designed and optimized **SQL-based ETL pipelines and data transformations** for telecom analytics datasets supporting finance reporting across **35+ markets**, improving query performance by **28%**.
- Improved accuracy of Vodafone UK's "**New Connection Indicator**" KPI from **63% to 92%** by auditing **data lineage, transformation logic, and validation rules** across production datasets.
- Built **analytics-ready data models and reporting pipelines** enabling near real-time KPI dashboards in Tableau while ensuring **data quality and reliability** across business systems.

IBM, *AI Developer Intern* | Global Delivery, Remote

Jul 2021 – Jun 2022

- Built **computer vision models using MediaPipe and OpenCV**, achieving **95% real-time gesture recognition accuracy**.
- Developed an **image classification pipeline using transfer learning (MobileNetV2)** achieving **98% model accuracy**.
- Deployed ML prototypes through **REST APIs and Node-RED workflows**, improving inference latency by **20%**.

PROJECTS

LLM – Powered Resume Matching & Candidate Scoring System | Ongoing (RAG, OpenAI/HuggingFace, FAISS, Python)

- Build an end-to-end LLM-based system to parse resumes and job descriptions, generating structured insights and **semantic embeddings** for candidate matching.
- Implement a **Retrieval-Augmented Generation (RAG)** pipeline using **vector search (FAISS)** to rank candidates based on contextual similarity and role requirements.
- Design workflows to process unstructured text data, perform information extraction, and automate candidate scoring using **prompt engineering** and transformer models.

AI-Powered Toxic Comment Detection and Moderation System | [GitHub](#) (PyTorch, HuggingFace, scikit-learn, Git LFS)

- Fine-tuned a **Transformer-based NLP classifier** using **PyTorch and HuggingFace** to detect multiple toxicity categories.
- Built an end-to-end **machine learning pipeline including preprocessing, training, evaluation, and inference workflows** on a **50K dataset**.
- Achieved **~96% accuracy and ~0.96 AUC**, validating performance using ROC curves, precision-recall metrics, and confusion matrices.

Transportation Data Science Project (TDSP) | [Certification](#) (Python, Pandas, Seaborn, Matplotlib, Folium)

- Analyzed NYC crash data to identify time-based risk patterns, including day-to-night transition accident spikes.
- Built interactive geospatial and statistical visualizations to communicate trends and support actionable safety recommendations.
- Shortlisted to present findings under the US DOT / Federal Highway Administration research certification program.

CERTIFICATIONS

- "Data Science Practitioner Certificate And Badge" by IBM